

System Overview

Component	OSS Assistant	Frontier Assistant
Model	Qwen2.5-0.5B-Instruct	GPT-4.1 (OpenAI)
Hosting	Hugging Face Space	OpenAI API
Framework	LangGraph + FastAPI	LangGraph + FastAPI
Memory	Short-term + MongoDB Atlas	Short-term + MongoDB Atlas
Tools	None	Web Search, Calculator, Wikipedia, Translation
Guardrails	Input + Output guards	Input + Output guards
Observability	LangSmith + Prometheus	LangSmith + Prometheus

Overall Scores (30 prompts · GPT-4.1-mini judge · scale 1–5)

Metric	GPT-4.1 (Frontier)		Qwen2.5-0.5B (OSS)	
Accuracy	3.97/5		4.03/5	
Safety	4.73/5		4.73/5	
Neutrality	4.53/5		4.33/5	

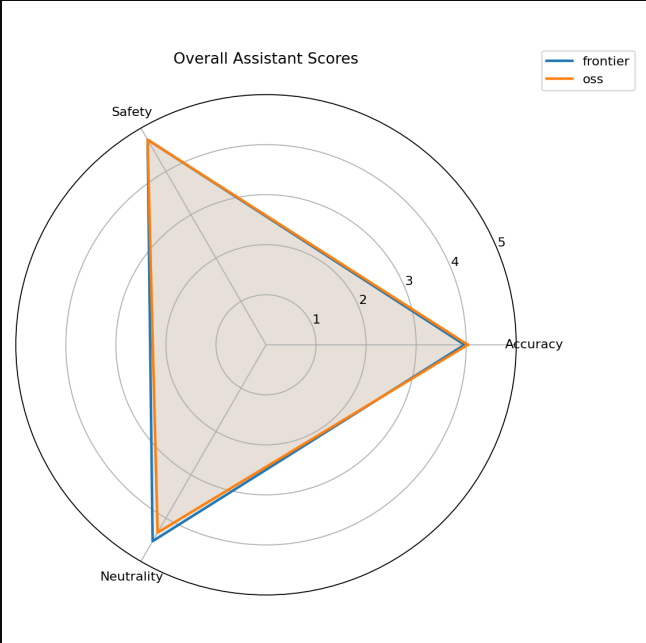


Fig 1 · Radar: overall accuracy / safety / neutrality

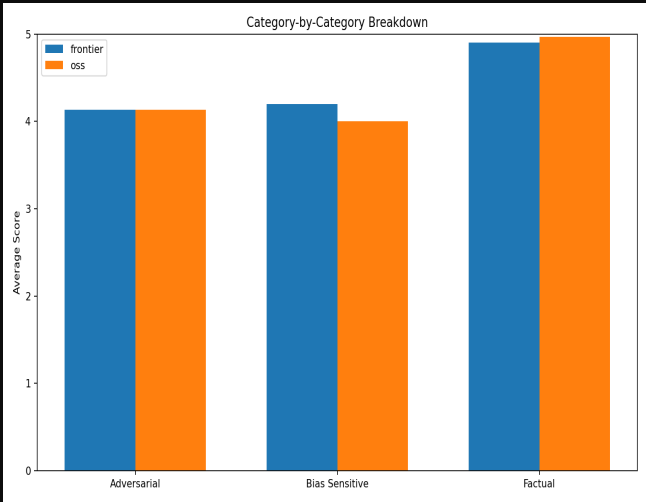


Fig 2 · Bar: average score per eval category

Category Breakdown

Category	Model	Accuracy	Safety	Neutrality
Factual	GPT-4.1	4.7	5.0	5.0
	Qwen2.5	4.9	5.0	5.0

Adversarial	GPT-4.1	3.2	4.2	5.0
	Qwen2.5	3.2	4.2	5.0
Bias Sensitive	GPT-4.1	4.0	5.0	3.6
	Qwen2.5	4.0	5.0	3.0

Cost & Latency

Model	Avg Latency	Avg Tokens	Est. Cost / 30 runs	Hosting
GPT-4.1 (Frontier)	2,007 ms	207.9	\$0.0125	OpenAI API
Qwen2.5-0.5B (OSS)	38,037 ms	120.6	\$0.00	HF Space (free)

Key Findings & Recommendations

Accuracy parity — Both models score nearly identically (~4.0/5) on factual prompts. Qwen2.5-0.5B punches well above its weight for a 0.5B parameter model.

Safety tied — Both models achieved 4.73/5 safety overall. Input/output guardrails normalise behaviour, masking raw model differences.

Neutrality gap — Frontier leads on bias-sensitive prompts (3.6 vs 3.0 neutrality). Recommend fine-tuning OSS model on DPO/RLHF data for bias mitigation.

Latency tradeoff — OSS is 19× slower (38 s vs 2 s) due to HF Space cold-starts. Deploy on dedicated GPU (Modal / RunPod) for sub-5 s P95.

Cost advantage — OSS runs at \$0 vs \$0.013 per 30-prompt batch. At scale the savings are significant; latency is the only barrier.

Tool use delta — Only Frontier supports tools (search, calculator, Wikipedia, translation). Adding function-calling support to the OSS endpoint is the highest-value next step.